

2021

YEAR 11 YEARLY EXAMINATION

Mathematics Standard

Examiner: Sami El Hosri

General Instructions

- Reading Time – 10 minutes
- Working Time – 2 hours
- Write using black pen
- Calculators approved by NESa may be used
- A separate reference sheet is provided at the back of this paper.
- All necessary working should be shown in Questions 16 – 36

Total marks – 75

Section I: Multiple Choice

Questions 1 – 15 15 marks

- Attempt all questions
- Allow about 25 minutes for this section

Section II: Extended Response

Questions 16 – 36 60 marks

- Attempt all questions
- Allow about 1 hour 35 minutes for this section

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Section I

Questions 1 – 15 (15 marks)

Read each question and choose an answer A, B, C or D.

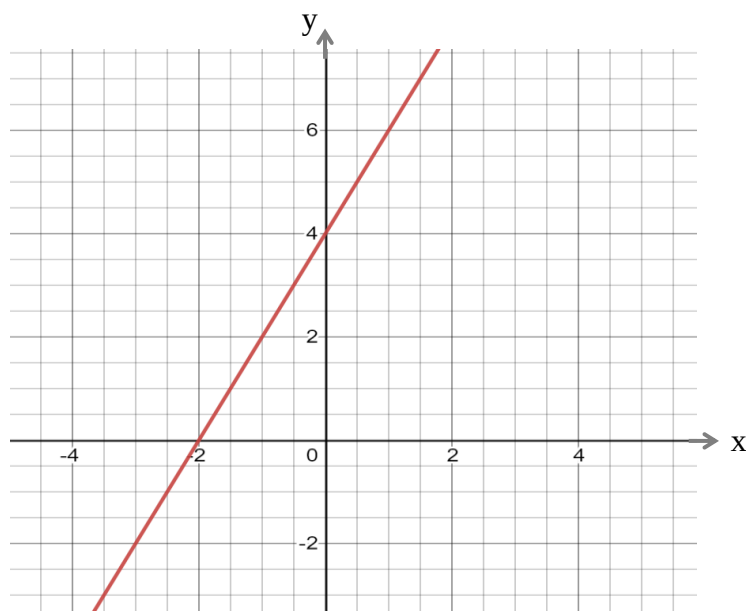
Allow about 25 minutes for this section.

Use the multiple-choice answer sheet for Questions 1-15.

1. Which of the following is an example of quantitative continuous data?

- A) The number of goals scored in a soccer game.
 - B) The number of aces scored by a tennis player in a game.
 - C) The time taken by a runner to complete a race.
 - D) The number of messages a person gets in a week.
-

2. What is the equation of the line shown in the number plane below?



- A) $y = 2x + 4$
 - B) $y = -2x + 4$
 - C) $y = 2x + 2$
 - D) $y = \frac{1}{2}x + 4$
-

3. David paid \$5 500 to buy a microwave, a fridge, and a TV.
He paid \$450 for the microwave and 60% of the remaining money for the fridge.
How much did he pay for the TV?

- A) \$3 030
 - B) \$2020
 - C) \$3 300
 - D) \$2 200
-

4. Mexico City in Mexico has a UTC of -6 and Sydney in Australia has a UTC of $+10$.

What is the day and time in Mexico City when it is 2 am Sunday in Sydney?

- A) 6 pm Sunday
 - B) 10 pm Saturday
 - C) 11 am Saturday
 - D) 10 am Saturday
-

5. Jacob earns \$650 per week plus \$75 for every 2 items he sells.

How many items did he sell in the last two weeks if his income was \$3100?

- A) 48
 - B) 24
 - C) 33
 - D) 64
-

6. Andrew is playing ten pin bowling. His average score in the first 5 games is 180 points. After playing 3 more games his overall average is now 183 points.

What is his average in the last three games?

- A) 187
 - B) 188
 - C) 190
 - D) 192
-

7. The height of a cylinder is 10 cm and its volume is $1\,392\text{ cm}^3$.

What is the radius of the cylinder correct to one decimal place?

- A) 6.6 cm
 - B) 43.3 cm
 - C) 6.7 cm
 - D) 7.0 cm
-

8. Melissa says the height of her cabinet is 200 cm, to the nearest centimetre.

What is the percentage error in her measurement?

- A) $\pm 0.025\%$
 - B) $\pm 0.5 \%$
 - C) $\pm 0.005\%$
 - D) $\pm 0.25\%$
-

9. The dosage of medication for children aged 1 to 2, based on an adult dosage, can be calculated using Fried's formula shown below.

$$\text{Child dosage} = \frac{\text{age of child (in months)} \times \text{adult dosage}}{150}$$

Using this formula, the dosage of a child is 3 mL based on an adult dosage of 25 mL.

How old is this child?

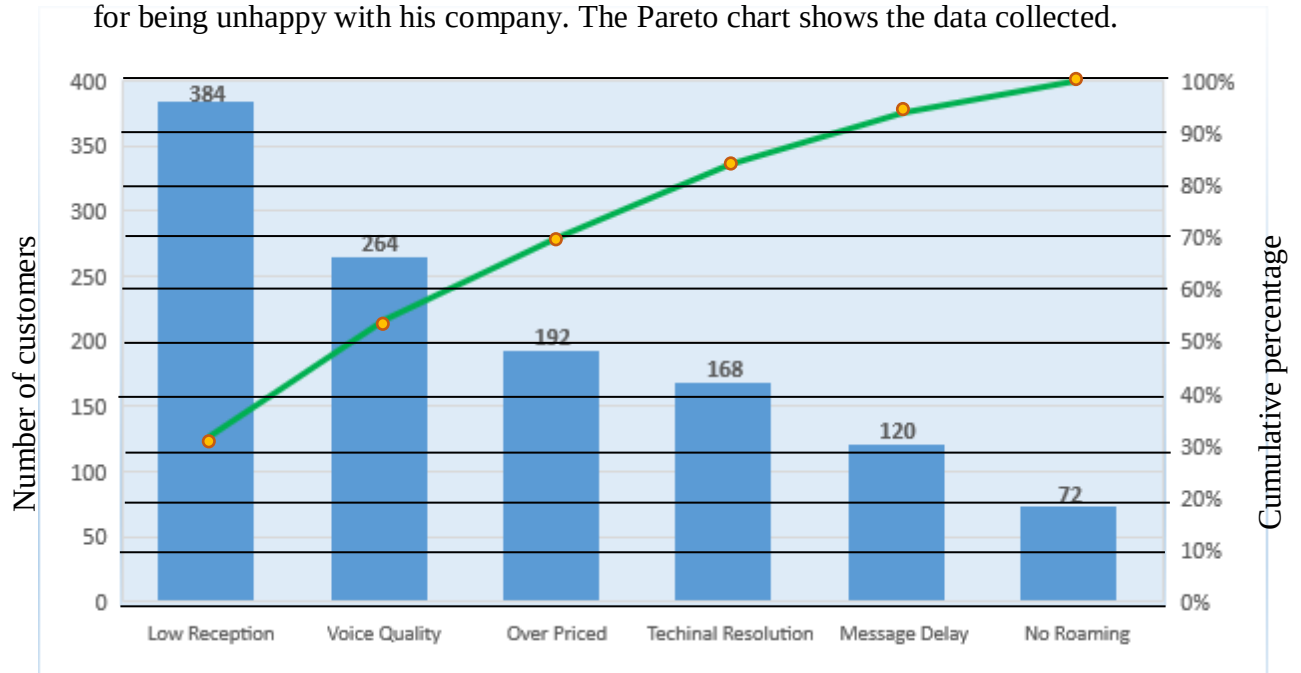
- A) 6 months
 - B) 9 months
 - C) 18 months
 - D) 20 months
-

10. A jar contains 27 lollies. 16 of these lollies are honey and 11 are strawberry. Andrew is to select 2 lollies from the jar at random.

What is the probability that Andrew selects two honey lollies?

- A) $\frac{16}{27} \times \frac{15}{26}$
 - B) $\frac{16}{27} + \frac{15}{27}$
 - C) $\frac{16}{27} + \frac{15}{26}$
 - D) $\frac{16}{27} \times \frac{15}{27}$
-

11. Clear Voice mobile manager collected data given by customers related to the reasons for being unhappy with his company. The Pareto chart shows the data collected.



What percentage of the customers were unhappy because the service was overpriced?

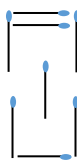
- A) 70%
 - B) 16%
 - C) 54%
 - D) 20%
-
12. David wants to buy a laptop for \$2 800. He already saved \$400 and each week, he plans to save \$120.

David follows his saving plan for t weeks, but he did not save enough to buy the laptop.

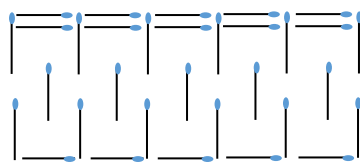
Which of the following equations shows the money M he still needs?

- A) $M = 2\,400 - 400t$
 - B) $M = 2\,400 + 120t$
 - C) $M = 2\,680 - 400t$
 - D) $M = 2\,400 - 120t$
-

13. A shape is created using 8 matches.



By adding more matches, we can create a row of five shapes as shown.



By continuing this pattern, what is the maximum number of shapes we can create using 200 matches?

- A) 33 B) 32 C) 34 D) 25
-
14. Students in a class sat for a physics test.

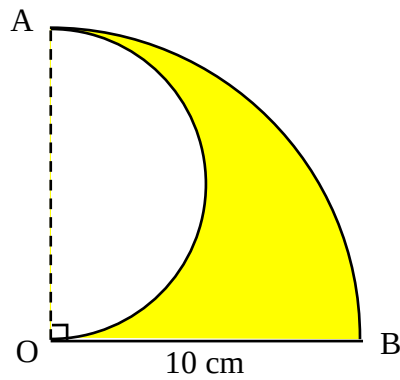
Because the test was considered too hard and the results were low, the teacher scaled the marks by multiplying each student's mark by 1.2.

The set of the students' scaled marks has a mean of 66 and standard deviation of 10.2.

What were the mean and the standard deviation of the original set of marks before scaling?

- A) The mean was 55 and the standard deviation was 9.
B) The mean was 66 and the standard deviation was 8.5.
C) The mean was 55 and the standard deviation was 8.5.
D) The mean was 55 and the standard deviation was 10.2.
-

15. A shape is formed by removing a semi-circle with diameter 10 cm from a quarter of a circle with radius 10 cm as shown in the diagram.



What is the area of the shaded shape correct to three significant figures?

- A) 58.9 cm^2
 - B) 39.3 cm^2
 - C) 78.5 cm^2
 - D) 275 cm^2
-

Year 11 MATHEMATICS STANDARD 2

Section II Answer Booklet

60 marks

Attempt Questions 16 – 36

Allow about 1 hour and 35 minutes for this section.

INSTRUCTIONS

- Answer the questions in the spaces provided. Sufficient space is provided for typical responses.
 - Your responses should include relevant mathematical reasoning and/or calculations.
 - Extra writing space is provided at the back of this booklet.
-

Question 16 (2 marks)

Robert surveyed seven people. He asked them how much they paid for water in their last bill.

Here are the results: \$350, \$470, \$530, \$540, \$380, \$480, \$460.

a) Calculate the mean of this set of data, correct to two decimal places.

1

b) What is the standard deviation of this set of data, correct to two decimal places?

1

Question 17 (2 marks)

An electrician charges a call-out fee of \$90 and then \$80 per hour while working. He also discounts 60% from the call-out fee if the job takes more than two hours.

2

How much will an electrician charge for a 3 hour job ?

Question 18 (2 marks)

How many kilocalories (kcal) are in a can of Best Salmon containing 12.2 grams of fat, correct to two decimal places?

2

Use the following conversions 1 gram of fat contains 37 kilojoules (kJ) and 1 kilocalorie = 4.184 kilojoules.

Question 19 (2 marks)

Draw a possible graph for a positively skewed distribution curve showing the possible location of the mode and the mean.

2

Question 20 (2 marks)

Amanda used a 2400 W washing machine for a total of 28 hours during the last 4 weeks.

2

What is the average cost of using this washing machine per week, if the cost of electricity is \$0.25 per kWh?

Question 21 (2 marks)

Mario's normal hourly rate is \$24.

2

On Saturdays he is paid \$80 for the first 2 hours and time and a half after that. Last Saturday he worked 8 hours.

How much did he earn for his work on that day?

Question 22 (3 marks)

Samantha wants to buy a boat for \$15 000. She decides to buy it on terms as follows:

3

- 20% deposit
- Balance and interest to be paid in equal monthly instalments of \$717 over 18 months.

What is the annual rate of simple interest that Samantha will pay on the money borrowed to buy the boat, correct to two decimal places?

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Question 23 (2 marks)

Joanne's monthly income is \$4 000. She is considering her monthly budget for the next 6 months to determine her total savings.

2

Her average expenses per month over a period of six months is \$2480.

How much will Joanne save in the next 6 months?

[illegible]

Question 24 (2 marks)

On Green Island, the goods and services tax GST is 12%.

Rebecca was on holiday on Green Island, and she bought a diamond ring for \$2 408 including GST.

When she was in the airport leaving the island, she received a refund for the GST paid on her diamond ring.

How much money did she receive?

2

Question 25 (3 marks)

Leader credit card charges interest at 18% per annum, compounded daily on any amount owing. Raymond used a Leader credit card to buy a laptop for \$2 800.

3

He owes no money on the credit card before purchasing the laptop.

Fifteen days later, he paid \$1150 off his credit card account.

How much will he still owe on the credit card immediately after his payment is made if he was charged interest daily, compounding for 15 days?

Question 26 (3 marks)

Data was collected from 16 real estates about the number of houses that they sold in the last six months. The set of data collected is displayed in the stem and leaf plot below.

Houses sold in last six months

1	5
2	6 6 9 9 9 9
3	0 2 2 4 6 8 8
4	2 4

a) What is the interquartile range of the data?

1

b) Is 15 an outlier for this set of data? Justify your answer using suitable calculations.

2

Question 27 (2 marks)

A patient received 180 mL of a medicine in 1 hour and 40 minutes through an intravenous drip at a constant rate.

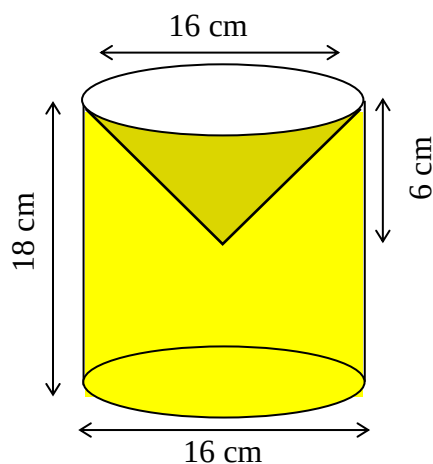
2

How long in hours and minutes will it take for the patient to receive a dose of 225 mL?

Question 28 (4 marks)

A solid is formed by removing a cone from a cylinder with diameter 16 cm and height is 18 cm.

The diameter of the cone is 16 cm, and its height is 6 cm as shown in the diagram below.



- a) Find the volume of the solid correct to two decimal places.

2

- b) Given that the curved area of the cone is $80\pi \text{ cm}^2$, find the surface area of the solid correct to two decimal places.

2

Question 29 (5 marks)

Last financial year Pamela's gross income was \$105 260, and her taxable income was \$101 936.

1

a) How much were her taxable deductions as a percentage of her gross income, correct to two decimal places?

b) Use the tax table below to calculate Pamela's tax on her taxable income.

2

Taxable income	Tax on this income
\$87,001 – \$180,000	\$19,822 plus 37c for each \$1 over \$87,000

c) Calculate Pamela's Medicare levy which is 2% of her taxable income.

1

d) After paying her calculated tax and her Medicare levy, Pamela received \$2 140 refund on her last financial year tax return from the tax office.

1

How much tax did she pay through her PAYG, throughout the year?

Question 30 (6 marks)

Brett went on holidays and had \$8 000. From the start of his holidays, he was spending his money at a constant rate of \$600 every 4 days.

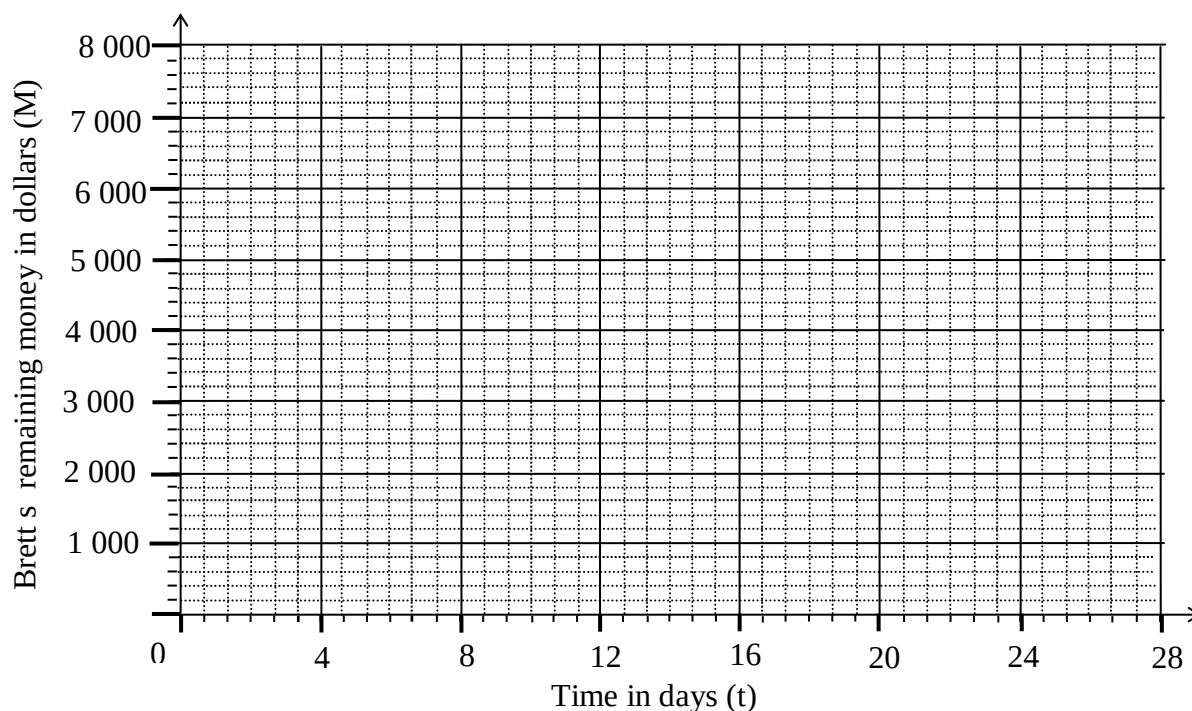
a) Use this information to complete the table below.

1

Time in days (t)	0	4	8	12	16
Brett's remaining money (M)	\$8 000	\$7 400		\$6 200	

b) Plot the ordered pairs from the table above to graph the line showing Brett's remaining money, in dollars, after t days.

1



c) Find the equation of Brett's remaining money in dollars, t days after he started his holidays.

2

d) After how many days will his remaining money be \$2600?

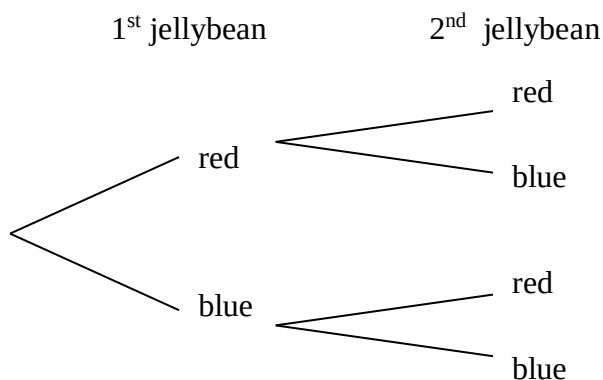
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Question 31 (2 marks)

A jar contains 8 red jellybeans and 6 blue jellybeans. Layla is to select two jellybeans from the jar at random.

- a) Complete the diagram by writing the probabilities on all the branches.

1



- b) Find the probability that Layla selects at least one red jellybean.

1

Question 32 (3 marks)

The energy used in kilojoules per kilogram of body mass, by a cyclist riding at an average speed of 12 km/h, is 10.54 kJ/kg per 30 minutes of ride.

3

Kevin ate 2 burgers each containing 150 kilocalories and drank a soft drink that contains 170 kilocalories.

Kevin's mass is 84 kg and he wants to ride his bike at an average speed of 12 km/h, until he burns off the energy he consumed.

How long, to the nearest minute, does Kevin need to ride his bike? (1 kilocalorie = 4.184 kJ)

Question 33 (2 marks)

A water tank contained 200 litres. During a storm, the tank was filled with storm water at a constant rate of 35.8 litres per minute.

2

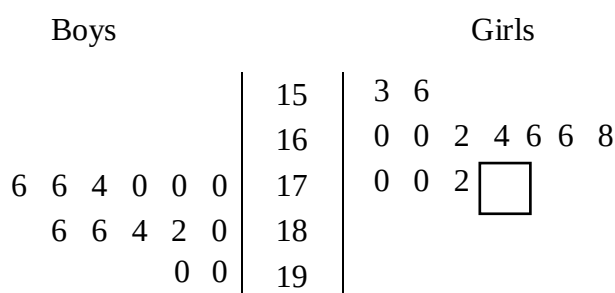
At the end of the storm, the tank contained 3 959 litres of water.

How long did the storm last?

Question 34 (3 marks)

The stem and leaf plot below shows the heights of a group of boys and a group of girls.

The height of Rebecca, the tallest in the girls' group, has a missing digit represented by .



It is known that the range of the girls' group is greater than the range of the boys' group.

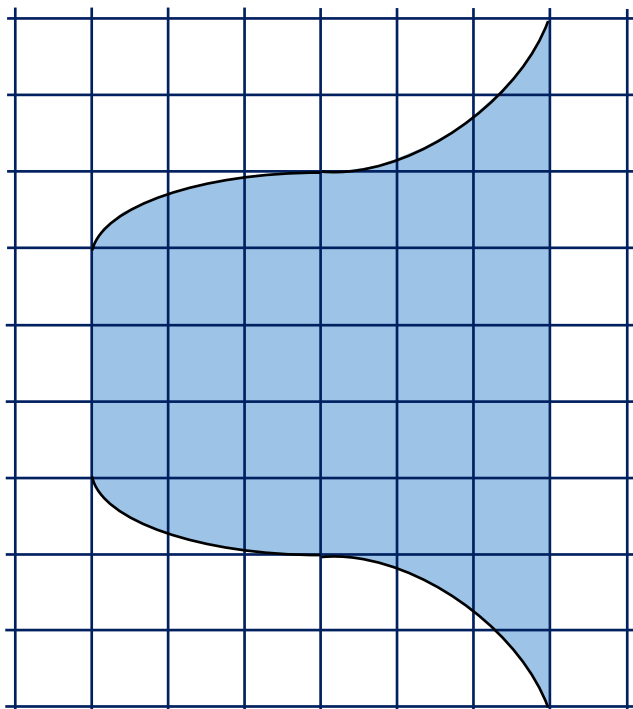
It is also known that the difference of the ranges of the two groups is equal to the difference of their interquartile ranges.

Calculate the value of the missing digit in Rebecca's height.

3

Question 35 (5 marks)

The shaded region on the diagram represents a swimming pool, viewed from above.
The scale is 1 cm = 3 m.



- a) Use two applications of the trapezoidal rule to calculate the approximate area of the top surface of the pool.

3

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- b) Clara states that the two applications of the trapezoidal rule will give the exact area of the pool. Do you agree or disagree with Clara's statement? Give mathematical reason.

2

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Question 36 (3 marks)

3

Steven weighs 80 kg. Last Saturday, p minutes after arriving at a party he started drinking alcohol. He stopped drinking after $4p$ minutes having consumed 2 bottles, each equivalent to 1.24 standard drinks.

He left the party at 1:00 a.m, 50 minutes after he stopped drinking, when his BAC had just reached zero.

Given that $BAC_{Male} = \frac{10 N - 7.5 H}{6.8 M}$, where N is the number of standard drinks consumed, H is the number of hours of drinking and M is a person's mass in kilograms.

Also, the time needed for BAC to reach zero after the last drink consumed can be calculated using $T = \frac{BAC}{0.015}$, where T is the time in hours.

At what time did Steven arrive at this party?

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END OF PAPER

Section II extra writing space

If you use this space, clearly indicate which question you are answering.

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REFERENCE SHEET

Measurement

Precision

Absolute error = $\frac{1}{2} \times \text{precision}$

Upper bound = measurement + absolute error

Lower bound = measurement – absolute error

Length, area, surface area and volume

$$l = \frac{\theta}{360} \times 2\pi r$$

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(x + y)$$

$$A \approx \frac{h}{2}(d_f + d_l)$$

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Trigonometry

$$A = \frac{1}{2}ab \sin C$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$c^2 = a^2 + b^2 - 2ab \cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

Financial Mathematics

$$FV = PV(1 + r)^n$$

Straight-line method of depreciation

$$S = V_0 - Dn$$

Declining-balance method of depreciation

$$S = V_0(1 - r)^n$$

Statistical Analysis

$$z = \frac{x - \bar{x}}{s}$$

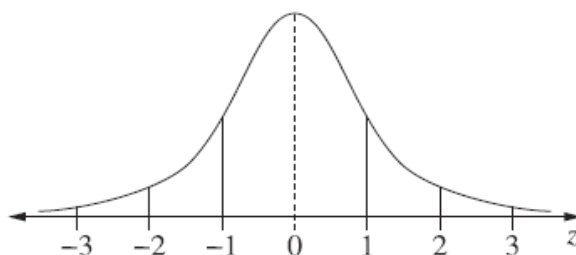
An outlier is a score

less than $Q_1 - 1.5 \times IQR$

or

more than $Q_3 + 1.5 \times IQR$

Normal distribution



- approximately 68% of scores have z -scores between -1 and 1
- approximately 95% of scores have z -scores between -2 and 2
- approximately 99.7% of scores have z -scores between -3 and 3